

1. The following table classifies the passengers on a flight from New York to Los Angeles based on vaccination status.

	Unvaccinated (0)	One Vaccine (1)	Two Vaccines (2)	Boosted (3)	Total
Age under 18 (A)	6	4	14	6	30
Age 18-30 (B)	21	8	34	15	78
Age 31-50 (C)	23	12	51	40	126
Age over 50 (D)	10	2	14	40	66
Total	60	26	113	101	300

Suppose that one passenger is selected at random from this flight.

- (2 points) Find the conditional probabilities $P(A|0)$ and $P(0|A)$.
 - (2 points) Find the probability that the passenger was vaccinated at least once.
 - (2 points) Find the probability that the passenger was at least 18 or unvaccinated.
 - (2 points) Using this data, give an example of two events that are mutually exclusive and justify.
 - (2 points) Are the events A and 0 independent? Justify your answer mathematically.
2. A cat sanctuary has at present a variety of cats taking refuge there. There are 10 black cats, 6 grey cats, 5 tabbies, and 4 calicos.
- (1 point) The veterinarian randomly collects five of the cats to give them check-ups. How many ways can she select this group?
 - (3 points) What is the probability that among the five cats selected, there are 3 black and 2 grey cats?
 - (3 points) If instead, the vet selects two cats, one at a time, what is the probability that the first is a tabby and the second is a calico?
3. At a recent classroom exam, 20% of the students finished the exam early (in the first hour). Among those who finished early, 40% passed the exam, whereas the pass rate for the class as a whole was 80%.
- (3 points) Joey was one of the students in the class who did not pass the exam. What is the probability that he finished early?
 - (3 points) Trudie was one of the students who did not finish early. What is the probability that she passed?
4. Consider the random variable X to be the number of absences per student in a certain course during one recent semester. *The teacher stopped counting once a student missed more than four classes.*

Number of absences (x)	0	1	2	3	4
Probability ($P(x)$)	0.82	0.08	0.05	??	0.01

- (1 point) Find the missing probability.
- (2 points) Calculate the expected number of absences per semester.
- (3 points) Calculate the standard deviation of X .
- (2 points) What is the probability of a student missing at most 2 classes in the semester?

5. Suppose that among one rare breed of dog, the Poopsidoodle, 75% will bark inside a house when the doorbell rings.
- (a) (3 points) In a random sample of 8 Poopsidoodles, calculate the probability that all 8 of them bark when the doorbell rings.
 - (b) (2 points) In a random sample of 8 Poopsidoodles, calculate the probability that at least 1 of them will **not** bark when the doorbell is rung.
6. The average adult human has an average attention span of 8 seconds with a standard deviation of 2 seconds. It was determined that the human attention span follows a normal distribution.
- (a) (3 points) Find the probability that a randomly selected individual has an attention span between 5 and 9 seconds.
 - (b) (4 points) Find the attention span in seconds representing that of the highest 5% of the population.
7. Suppose that a typical resident of Québec enjoys eating 12 poutines a month with a standard deviation of 5 poutines.
- (a) (4 points) What is the probability that the average number of poutines consumed by a sample of 45 Québécois would be greater than 13 poutines?
 - (b) (1 point) If the sample size was to be increased to 90 would the probability found in part (a) increase, decrease or stay the same and why?
8. About 14% of Ontarian high-school students experience severe psychological distress (symptoms of anxiety or depression) at some point during their pre-university education.
- (a) (2 points) If a class of high-school students in Ontario is randomly selected, what is the minimum class size required so that the normal distribution could be used as a valid approximation for calculating probabilities?
 - (b) (4 points) In a classroom of 40 students, what is the **approximate** probability that less than 8 of them experienced severe psychological distress during highschool?
9. At a large tech company, *Bananas*, the number of vacation days for employees follows an unknown distribution. You, a curious and nosey employee, want to estimate the average number of vacation days your company gives to their employees.
- (a) (3 points) Stella from H.R. tells you that population standard deviation for the number of vacation days is 3. Find the smallest number of employees who you would need to sample so as to ensure with 95% confidence, that your sample mean will be within 0.5 days of the true population mean.
 - (b) (4 points) After conducting the survey with a sample size of $n = 150$ employees, you obtain a point estimate of 15 days. Construct a 90% confidence interval estimating the true average number of vacation days for all employees at *Bananas*.
 - (c) (4 points) You suspect that the standard deviation that Stella provided is inaccurate so you decide to instead use your sample standard deviation $s = 5$. Using $n = 150$, construct a new 90% confidence interval to estimate the true average number of vacation days for all employees at *Bananas*.

10. According to a survey the average age of crafters (knitting, crochet, sewing, ...) is 49.1. A random sample of 48 crafters at the Woolly festival gave a sample average of 45.5 years. Suppose that we know that the population standard deviation for the average age of crafters is 14.3 years. Test the hypothesis that the actual average age of crafters is less than 49.1 years, using a 1% significance level.
- (2 points) State the null and alternative hypotheses.
 - (3 points) Calculate the test statistic.
 - (3 points) Find (or estimate) the p -value for the test.
 - (2 points) State and **interpret** your conclusion.
 - (1 point) Would your conclusion change if we used a 5% significance level? Explain.
11. The following exercise is based on *A ventilation intervention study in classrooms to improve indoor air quality: the FRESH study (2013, The Netherlands)*.

Classroom ventilation rates often do not meet building standards, although it is considered to be important to improve indoor air quality. Poor indoor air quality is thought to influence both children's health and performance. Poor ventilation most often occurs in the heating season. To improve classroom ventilation a tailor made mechanical ventilation device was developed to increase outdoor air supply.

CO₂ concentration measurements inside the classrooms were used to determine indoor air quality. During week A mechanical ventilation was set to start when CO₂ concentration reaches 800 ppm (parts per million). During week B it was set to start at 1200 ppm. The below table gives the mean indoor CO₂ concentration (ppm) per school per week.

School	Ventilation at 1200 ppm (week B)	800 ppm (week A)	$d = B - A$	$(d - \bar{d})^2$
1	900	900	0	16384
2	910	800	110	324
3	900	750	150	484
4	1060	840	220	8464
5	1070	910	160	1024
			$\sum d = 640$	$\sum (d - \bar{d})^2 = 26680$

Does this data indicate that setting the ventilation to start at 800 ppm was more effective in reducing mean indoor CO₂ concentration than at 1200 ppm? Use a 5% significance level.

- (2 points) State the null and alternative hypotheses.
- (3 points) Calculate the test statistic.
- (3 points) Find (or estimate) the p -value for the test.
- (2 points) State and **interpret** your conclusion.

12. About 33% of Quebecers claim that they cooked more often during the pandemic. Suppose a random sample of 162 students were surveyed, and 68 of them said that they cooked more often during the pandemic. Test the hypothesis that the proportion of students who cooked more often during the pandemic is different from that of the general population. Use a 5% significance level.
- (a) (2 points) State the null and alternative hypotheses.
 - (b) (3 points) Calculate the test statistic.
 - (c) (3 points) Find (or estimate) the p -value for the test.
 - (d) (2 points) State and **interpret** your conclusion.
13. (4 points) Swimming is a popular activity for the treatment and prevention of lower back pain. 85 participants in an experiment were randomly divided into two groups. The first group (43 people) received only educational materials, whereas the second group (42 people) participated in 6 weeks of swimming lessons. At a 6-week follow-up the sample mean level of pain (on a scale from 0 to 10) for the first group was 5.1 with sample standard deviation 2.5 and the sample mean for the second group was 3.5 with sample standard deviation 2.7.
- Let μ_1 = the population mean level of pain of people who do *not* swim regularly and let μ_2 = the population mean level of pain of people who do.
- Find a 90% confidence interval for $\mu_1 - \mu_2$.

ANSWERS

1. (a) $P(A|0) = \frac{6}{60} = 0.100$
 $P(0|A) = \frac{6}{30} = 0.200$.
 (b) $\frac{240}{300} = 0.800$
 (c) $\frac{276}{300} = 0.920$
 (d) Answers vary. One possibility:
 The events 0 and 1, since $P(0 \text{ and } 1) = 0$.
 (e) A and 0 are independent since
 $P(A) = \frac{30}{300} = 0.100 = P(A|0)$
2. (a) $25C5 = 53130$
 (b) $\frac{10C3 \cdot 6C2}{25C5} = \frac{1800}{53130} = 0.0339$
 (c) $\frac{5}{25} \cdot \frac{4}{24} = \frac{20}{600} = 0.0333$
3. (a) $P(\text{early}|\text{fail}) = \frac{12}{20} = 0.600$
 (b) $P(\text{pass}|\text{late}) = \frac{72}{80} = 0.900$
4. (a) 0.04 (c) 0.8273
 (b) 0.34 (d) 0.95
5. (a) 0.1001
 (b) 0.8999
6. (a) 0.6247
 (b) 11.29 seconds
7. (a) 0.0901
 (b) The probability would decrease since the larger sample size would decrease the standard deviation of the normal distribution, creating a narrower bell curve. The area in the tail to the right of 13 poutines would thus be reduced. Alternatively, you could argue that it is less likely to find a larger sample of people who eat on average so many poutines each month.
8. (a) $n = 36$ so that $n(0.14) > 5$ & $n(0.86) > 5$
 (b) 0.8078
9. (a) $n = 139$
 (b) [14.597, 15.403]
 (c) [14.322, 15.678]
10. (a) $H_0 : \mu = 49.1$; $H_a : \mu < 49.1$
 (b) $z^* = -1.74$
 (c) p-value = 0.0409
 (d) Fail to reject H_0 , There is insufficient evidence to support the claim that the average age of crafters is less than 49.1 years.
 (e) Yes, the p-value would be less than α
11. (a) $H_0 : \mu_d = 0$; $H_a : \mu_d > 0$
 (b) $t^* = 3.5045$
 (c) $0.01 < \text{p-value} < 0.025$
 (d) Reject H_0 , There is sufficient evidence to support the claim that setting the ventilation to start at 800 ppm was more effective.
12. (a) $H_0 : p = 0.33$; $H_a : p \neq 0.33$
 (b) $z^* = 2.43$
 (c) p-value = 0.015
 (d) Reject H_0 , There is sufficient evidence to support the claim that the proportion of students who cooked during the pandemic is different from that of the general population.
13. (0.6490, 2.5510)